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Last login: Sun Jun 28 06:13:31 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Torus"
echo "==== folder contents ====="
ls -la
echo ""
echo "==== PPN=1 torus config ====="
cat torus_ppn1.py 2>/dev/null || { echo "torus_ppn1.py not found here - listing .py files:"; ls *.py 2>/dev/null; }
find . -name "*ppn1*.py" 2>/dev/null; }
==== folder contents =====
total 2984
drwxr-xr-x 23 stephenrajva staff 736 24 Jun 19:14 .
drwxr-xr-x 9 stephenrajva staff 288 24 Jun 23:05 ..
-rw-r--r--@ 1 stephenrajva staff 10244 25 Jun 00:49 .DS_Store
-rw-r--r--@ 1 stephenrajva staff 6972 24 Jun 18:54 analyze_universal.py
-rw-r--r-- 1 stephenrajva staff 1534 18 Jun 20:51 extract_torus_metrics.py
-rw-r--r-- 1 stephenrajva staff 35426 18 Jun 13:44 miniFE_communication_stats_torus.csv
-rw-r--r-- 1 stephenrajva staff 142042 18 Jun 20:41 miniFE_communication_stats_torus16.csv
-rw-r--r-- 1 stephenrajva staff 573512 18 Jun 20:50 miniFE_communication_stats_torus64.csv
-rw-r--r-- 1 stephenrajva staff 50828 24 Jun 18:54 MiniFE_Torus_bits.png
-rw-r--r-- 1 stephenrajva staff 49369 24 Jun 18:54 MiniFE_Torus_hops.png
-rw-r--r-- 1 stephenrajva staff 51136 24 Jun 18:54 MiniFE_Torus_idle_pct.png
-rw-r--r-- 1 stephenrajva staff 58733 24 Jun 18:54 MiniFE_Torus_latency_ns.png
-rw-r--r-- 1 stephenrajva staff 50595 24 Jun 18:54 MiniFE_Torus_packets.png
-rw-r--r-- 1 stephenrajva staff 32078 24 Jun 18:54 MiniFE_Torus_stalls.png
-rw-r--r-- 1 stephenrajva staff 49716 24 Jun 18:54 MiniFE_Torus_throughput.png
drwxr-xr-x 12 stephenrajva staff 384 24 Jun 18:45 results
-rwxr-xr-x@ 1 stephenrajva staff 487 24 Jun 18:44 run_torus_sweep.sh
-rw-r--r--@ 1 stephenrajva staff 357742 24 Jun 19:14 Screenshot 2026-06-24 at 7.14.09 PM.png
-rw-r--r--@ 1 stephenrajva staff 3884 24 Jun 18:44 torus_ppn1.py
-rw-r--r-- 1 stephenrajva staff 284 18 Jun 20:51 torus_summary.csv
-rw-r--r-- 1 stephenrajva staff 2535 18 Jun 20:39 torus16.py
-rw-r--r-- 1 stephenrajva staff 2538 18 Jun 13:44 torus4.py
-rw-r--r-- 1 stephenrajva staff 2535 18 Jun 20:50 torus64.py

==== PPN=1 torus config =====
#!/usr/bin/env python
# =====
# torus_ppn1.py - MiniFE on a Torus, PPN (ranks per node) = 1
# =====
# Same study as the MiniFE mesh, but topoTorus (wraparound links).
# PPN = ranks / nodes = size / (routers * local_ports). We set size = nodes,
# so PPN is ALWAYS 1.
#
# USAGE: export MESH_SHAPE="4x4" (then) sst torus_ppn1.py
# The runner loops over all shapes.
# =====

import sst, os
from sst.merlin.base import *
from sst.merlin.endpoint import *
from sst.merlin.interface import *
from sst.merlin.topology import *
from sst.ember import *

PPN = 1
shape = os.environ.get("MESH_SHAPE", "4x4")
nodes_per_router = 1
out_dir = os.environ.get("STATS_DIR", ".")

dims = [int(x) for x in shape.lower().split("x")]
num_routers = 1
for d in dims:
    num_routers *= d
num_nodes = num_routers * nodes_per_router
num_ranks = num_nodes * PPN # PPN=1 -> ranks == nodes

PlatformDefinition.setCurrentPlatform("firefly-defaults")

topo = topoTorus()
topo.shape = shape
topo.local_ports = nodes_per_router
topo.width = "x".join(["1"] * len(dims)) # 1 link per dim (matches mesh study)
topo.link_latency = "25ns"

router = hr_router()
router.link_bw = "12GB/s"
router.flit_size = "16B"
router.xbar_bw = "20GB/s"
router.input_latency = "30ns"
router.output_latency = "30ns"
router.input_buf_size = "32kB"

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router.output_buf_size = "32kB"
router.num_vns         = 2
router.xbar_arb         = "merlin.xbar_arb_lru"
topo.router = router

nic                     = ReorderLinkControl()
nic.link_bw             = "12GB/s"
nic.input_buf_size     = "16kB"
nic.output_buf_size    = "16kB"

system = System()
system.setTopology(topo)

# MiniFE motifs (identical to the mesh study so the comparison is fair)
motifs = [
    {"pattern": "Allreduce", "params": "iterations=92 count=1 compute=69ns"},
    {"pattern": "Bcast",      "params": "root=0 iterations=2 count=7 compute=1323ns"},
    {"pattern": "Reduce",     "params": "root=0 iterations=2 count=2 compute=450ns"},
    {"pattern": "Allgather",  "params": "iterations=3 count=3 compute=243ns verify=0"},
    {"pattern": "Barrier",    "params": "iterations=99 compute=75ns"},
]

row_step = dims[-1] if len(dims) >= 2 else 1
candidates = [1, row_step, num_nodes // 2, num_nodes - 1]
partners = sorted([p for p in candidates if 0 < p < num_ranks])
for b in partners:
    motifs.append({"pattern": "PingPong",
                  "params": f"rank2={b} iterations=35 messageSize=256 compute=3ns"})

ep = EmberMPIJob(0, num_ranks)
ep.network_interface = nic
ep.addMotif("Init")
for m in motifs:
    ep.addMotif(f"{m['pattern']} {m['params']}")
ep.addMotif("Fini")
ep.nic.nic2host_lat = "100ns"

system.allocateNodes(ep, "linear")
system.build()

csv_path = os.path.join(out_dir, f"torus_{shape}_ppn{PPN}_nodes{num_nodes}_stats.csv")
sst.setStatisticLoadLevel(10)
sst.setStatisticOutput("sst.statOutputCSV", {"filepath": csv_path, "separator": ", "})
sst.enableAllStatisticsForComponentType("merlin.hr_router")
sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
sst.enableAllStatisticsForComponentType("ember.EmberEngine", {"MotifLog": "1"})
sst.enableAllStatisticsForComponentType("ember.nic")

print("=====")
print(f" Topology      : Torus {shape}")
print(f" Routers       : {num_routers}")
print(f" Total nodes   : {num_nodes}")
print(f" Total MPI ranks : {num_ranks}")
print(f" PPN (ranks/node) : {num_ranks // num_nodes} <-- should be 1")
print(f" Stats CSV     : {csv_path}")
print("=====")
stephenrajva@Stephens-MacBook-Air-4 Torus % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST-Stimulator/MiniFE
/Torus/PPN_2"
chmod +x run_torus_sweep_ppn2.sh
export MESH_SHAPE="2x2"
export STATS_DIR="./results"
sst torus_ppn2.py
cd: no such file or directory: /Users/stephenrajva/Downloads/HPC-Upload 2/SST-Stimulator/MiniFE/Torus/PPN_2
chmod: run_torus_sweep_ppn2.sh: No such file or directory
Failed to canonicalize path [torus_ppn2.py]: No such file or directory
stephenrajva@Stephens-MacBook-Air-4 Torus % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST-Stimulator/MiniFE
/Torus"
mkdir -p PPN_2
echo "==== is PPN_2 there now? ====="
ls -ld PPN_2
echo "==== where are the downloaded files? ====="
ls -l ~/Downloads/torus_ppn2.py ~/Downloads/run_torus_sweep_ppn2.sh 2>/dev/null
echo "==== anything already in PPN_2? ====="
ls -l PPN_2/
==== is PPN_2 there now? =====
drwxr-xr-x  2 stephenrajva  staff   64 28 Jun 06:30 PPN_2
==== where are the downloaded files? =====
-rw-r--r--@ 1 stephenrajva  staff   524 28 Jun 06:29 /Users/stephenrajva/Downloads/run_torus_sweep_ppn2.sh
-rw-r--r--@ 1 stephenrajva  staff  4261 28 Jun 06:29 /Users/stephenrajva/Downloads/torus_ppn2.py
==== anything already in PPN_2? =====
total 0
stephenrajva@Stephens-MacBook-Air-4 Torus % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST-Stimulator/MiniFE
/Torus"

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mv ~/Downloads/torus_ppn2.py ~/Downloads/run_torus_sweep_ppn2.sh PPN_2/
ls -l PPN_2/
total 24
-rw-r--r--@ 1 stephenrajva  staff   524 28 Jun 06:29 run_torus_sweep_ppn2.sh
-rw-r--r--@ 1 stephenrajva  staff  4261 28 Jun 06:29 torus_ppn2.py
stephenrajva@Stephens-MacBook-Air-4 Torus % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE
/Torus/PPN_2"
chmod +x run_torus_sweep_ppn2.sh
export MESH_SHAPE="2x2"
export STATS_DIR="./results"
sst torus_ppn2.py
=====
Topology          : Torus 2x2
Routers           : 4
Nodes per router  : 2 (local_ports)
Total nodes       : 8
Total MPI ranks   : 8
PPN (ranks/node)  : 2  <-- should be 2
Stats CSV         : ./results/torus_2x2_ppn2_nodes8_stats.csv
=====
Allreduce: ranks 8, loop 92, 1 double(s), latency 5.507 us
generate
510120733:0:0:EmberEngine:Reduce:generate:37: rank=0 size=8
generate
511463933:0:1:EmberEngine:Reduce:generate:37: rank=1 size=8
generate
511679933:0:2:EmberEngine:Reduce:generate:37: rank=2 size=8
generate
512263133:0:3:EmberEngine:Reduce:generate:37: rank=3 size=8
generate
512263533:0:4:EmberEngine:Reduce:generate:37: rank=4 size=8
generate
512435533:0:5:EmberEngine:Reduce:generate:37: rank=5 size=8
generate
512479533:0:6:EmberEngine:Reduce:generate:37: rank=6 size=8
generate
Bcast: ranks 8, loop 2, bytes 56, latency 1.893 us
513062733:0:7:EmberEngine:Reduce:generate:37: rank=7 size=8
Allgather: ranks 8, loop 3, bytes 12, latency 5.041 us
Barrier: ranks 8, loop 99, latency 5.506 us
1099539533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=1 messageSize=256 iterations=35
1100382733:0:1:EmberEngine:PingPong:generate:65: rank=1 size=8
1100598733:0:2:EmberEngine:PingPong:generate:65: rank=2 size=8
1101181933:0:4:EmberEngine:PingPong:generate:65: rank=4 size=8
1101981933:0:7:EmberEngine:PingPong:generate:65: rank=7 size=8
PingPong: otherRank 1, total time 63.474 us, loop 35, buflen 256, latency 0.907 us. bandwidth 0.282320 GB/s
1163013133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=2 messageSize=256 iterations=35
PingPong: otherRank 2, total time 69.272 us, loop 35, buflen 256, latency 0.990 us. bandwidth 0.258690 GB/s
1232285133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=4 messageSize=256 iterations=35
PingPong: otherRank 4, total time 69.272 us, loop 35, buflen 256, latency 0.990 us. bandwidth 0.258690 GB/s
1301557133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=7 messageSize=256 iterations=35
PingPong: otherRank 7, total time 75.320 us, loop 35, buflen 256, latency 1.076 us. bandwidth 0.237918 GB/s
Simulation is complete, simulated time: 1.38146 ms
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE
/Torus/PPN_2"
./run_torus_sweep_ppn2.sh
=== MiniFE torus 2x2 (PPN=2) ===
=====
Topology          : Torus 2x2
Routers           : 4
Nodes per router  : 2 (local_ports)
Total nodes       : 8
Total MPI ranks   : 8
PPN (ranks/node)  : 2  <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Torus/PPN_2/results/torus_
2x2_ppn2_nodes8_stats.csv
=====
Allreduce: ranks 8, loop 92, 1 double(s), latency 5.507 us
generate
510120733:0:0:EmberEngine:Reduce:generate:37: rank=0 size=8
generate
511463933:0:1:EmberEngine:Reduce:generate:37: rank=1 size=8
generate
511679933:0:2:EmberEngine:Reduce:generate:37: rank=2 size=8
generate
512263133:0:3:EmberEngine:Reduce:generate:37: rank=3 size=8
generate
512263533:0:4:EmberEngine:Reduce:generate:37: rank=4 size=8

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generate
512435533:0:5:EmberEngine:Reduce:generate:37: rank=5 size=8
generate
512479533:0:6:EmberEngine:Reduce:generate:37: rank=6 size=8
generate
Bcast: ranks 8, loop 2, bytes 56, latency 1.893 us
513062733:0:7:EmberEngine:Reduce:generate:37: rank=7 size=8
Allgather: ranks 8, loop 3, bytes 12, latency 5.041 us
Barrier: ranks 8, loop 99, latency 5.506 us
1099539533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=1 messageSize=256 iterations=35
1100382733:0:1:EmberEngine:PingPong:generate:65: rank=1 size=8
1100598733:0:2:EmberEngine:PingPong:generate:65: rank=2 size=8
1101181933:0:4:EmberEngine:PingPong:generate:65: rank=4 size=8
1101981933:0:7:EmberEngine:PingPong:generate:65: rank=7 size=8
PingPong: otherRank 1, total time 63.474 us, loop 35, buflen 256, latency 0.907 us. bandwidth 0.282320 GB/s
1163013133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=2 messageSize=256 iterations=35
PingPong: otherRank 2, total time 69.272 us, loop 35, buflen 256, latency 0.990 us. bandwidth 0.258690 GB/s
1232285133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=4 messageSize=256 iterations=35
PingPong: otherRank 4, total time 69.272 us, loop 35, buflen 256, latency 0.990 us. bandwidth 0.258690 GB/s
1301557133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=7 messageSize=256 iterations=35
PingPong: otherRank 7, total time 75.320 us, loop 35, buflen 256, latency 1.076 us. bandwidth 0.237918 GB/s
Simulation is complete, simulated time: 1.38146 ms
=== MiniFE torus 4x4 (PPN=2) ===
=====
Topology      : Torus 4x4
Routers       : 16
Nodes per router : 2 (local_ports)
Total nodes    : 32
Total MPI ranks : 32
PPN (ranks/node) : 2 <-- should be 2
Stats CSV      : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Torus/PPN_2/results/torus_
4x4_ppn2_nodes32_stats.csv
=====
Allreduce: ranks 32, loop 92, 1 double(s), latency 9.586 us
generate
885339933:0:0:EmberEngine:Reduce:generate:37: rank=0 size=32
generate
886683133:0:1:EmberEngine:Reduce:generate:37: rank=1 size=32
generate
886899133:0:2:EmberEngine:Reduce:generate:37: rank=2 size=32
generate
887612733:0:3:EmberEngine:Reduce:generate:37: rank=3 size=32
generate
887828733:0:5:EmberEngine:Reduce:generate:37: rank=5 size=32
generate
887838133:0:4:EmberEngine:Reduce:generate:37: rank=4 size=32
generate
888045333:0:6:EmberEngine:Reduce:generate:37: rank=6 size=32
generate
888628733:0:7:EmberEngine:Reduce:generate:37: rank=7 size=32
generate
888758933:0:8:EmberEngine:Reduce:generate:37: rank=8 size=32
generate
888844733:0:11:EmberEngine:Reduce:generate:37: rank=11 size=32
generate
888888533:0:12:EmberEngine:Reduce:generate:37: rank=12 size=32
generate
888940533:0:9:EmberEngine:Reduce:generate:37: rank=9 size=32
generate
888984333:0:10:EmberEngine:Reduce:generate:37: rank=10 size=32
generate
889061333:0:13:EmberEngine:Reduce:generate:37: rank=13 size=32
generate
889105133:0:14:EmberEngine:Reduce:generate:37: rank=14 size=32
generate
889427933:0:15:EmberEngine:Reduce:generate:37: rank=15 size=32
generate
889428333:0:17:EmberEngine:Reduce:generate:37: rank=17 size=32
generate
889601133:0:16:EmberEngine:Reduce:generate:37: rank=16 size=32
generate
889645133:0:18:EmberEngine:Reduce:generate:37: rank=18 size=32
generate
889686733:0:23:EmberEngine:Reduce:generate:37: rank=23 size=32
generate
889696333:0:19:EmberEngine:Reduce:generate:37: rank=19 size=32
generate
889740333:0:21:EmberEngine:Reduce:generate:37: rank=21 size=32

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generate
889817133:0:24:EmberEngine:Reduce:generate:37: rank=24 size=32
generate
889826733:0:25:EmberEngine:Reduce:generate:37: rank=25 size=32
generate
889861133:0:26:EmberEngine:Reduce:generate:37: rank=26 size=32
generate
889903533:0:27:EmberEngine:Reduce:generate:37: rank=27 size=32
generate
889913133:0:20:EmberEngine:Reduce:generate:37: rank=20 size=32
generate
889947533:0:29:EmberEngine:Reduce:generate:37: rank=29 size=32
generate
889956333:0:22:EmberEngine:Reduce:generate:37: rank=22 size=32
generate
889957133:0:28:EmberEngine:Reduce:generate:37: rank=28 size=32
generate
889990733:0:30:EmberEngine:Reduce:generate:37: rank=30 size=32
generate
Bcast: ranks 32, loop 2, bytes 56, latency 1.893 us
890313933:0:31:EmberEngine:Reduce:generate:37: rank=31 size=32
Allgather: ranks 32, loop 3, bytes 12, latency 7.686 us
Barrier: ranks 32, loop 99, latency 9.580 us
1890847533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=1 messageSize=256 iterations=35
1891690733:0:1:EmberEngine:PingPong:generate:65: rank=1 size=32
1892840333:0:4:EmberEngine:PingPong:generate:65: rank=4 size=32
1894608333:0:16:EmberEngine:PingPong:generate:65: rank=16 size=32
1895321933:0:31:EmberEngine:PingPong:generate:65: rank=31 size=32
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.282320 GB/s
1954321133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=4 messageSize=256 iterations=35
PingPong: otherRank 4, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918 GB/s
2029641133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=16 messageSize=256 iterations=35
PingPong: otherRank 16, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918 GB/s
2104961133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=31 messageSize=256 iterations=35
PingPong: otherRank 31, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918 GB/s
Simulation is complete, simulated time: 2.18892 ms
=== MiniFE torus 8x8 (PPN=2) ===
=====
Topology      : Torus 8x8
Routers       : 64
Nodes per router : 2 (local_ports)
Total nodes    : 128
Total MPI ranks : 128
PPN (ranks/node) : 2 <-- should be 2
Stats CSV      : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Torus/PPN_2/results/torus_
8x8_ppn2_nodes128_stats.csv
=====
Allreduce: ranks 128, loop 92, 1 double(s), latency 14.883 us
generate
1372692733:0:0:EmberEngine:Reduce:generate:37: rank=0 size=128
generate
1374035933:0:1:EmberEngine:Reduce:generate:37: rank=1 size=128
generate
1374251933:0:2:EmberEngine:Reduce:generate:37: rank=2 size=128
generate
1374965533:0:3:EmberEngine:Reduce:generate:37: rank=3 size=128
generate
1375181533:0:5:EmberEngine:Reduce:generate:37: rank=5 size=128
generate
1375190933:0:4:EmberEngine:Reduce:generate:37: rank=4 size=128
generate
1375398133:0:6:EmberEngine:Reduce:generate:37: rank=6 size=128
generate
1375981533:0:7:EmberEngine:Reduce:generate:37: rank=7 size=128
generate
1376206933:0:8:EmberEngine:Reduce:generate:37: rank=8 size=128
generate
1376216533:0:9:EmberEngine:Reduce:generate:37: rank=9 size=128
generate
1376283933:0:11:EmberEngine:Reduce:generate:37: rank=11 size=128
generate
1376433333:0:10:EmberEngine:Reduce:generate:37: rank=10 size=128
generate
1376500533:0:13:EmberEngine:Reduce:generate:37: rank=13 size=128
generate
1376510133:0:12:EmberEngine:Reduce:generate:37: rank=12 size=128
generate
1376717133:0:14:EmberEngine:Reduce:generate:37: rank=14 size=128

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generate
1377170333:0:15:EmberEngine:Reduce:generate:37: rank=15 size=128
generate
1377300533:0:16:EmberEngine:Reduce:generate:37: rank=16 size=128
generate
1377386333:0:23:EmberEngine:Reduce:generate:37: rank=23 size=128
generate
1377405333:0:19:EmberEngine:Reduce:generate:37: rank=19 size=128
generate
1377430133:0:24:EmberEngine:Reduce:generate:37: rank=24 size=128
generate
1377449133:0:20:EmberEngine:Reduce:generate:37: rank=20 size=128
generate
1377482133:0:17:EmberEngine:Reduce:generate:37: rank=17 size=128
generate
1377516533:0:27:EmberEngine:Reduce:generate:37: rank=27 size=128
generate
1377525933:0:18:EmberEngine:Reduce:generate:37: rank=18 size=128
generate
1377560333:0:28:EmberEngine:Reduce:generate:37: rank=28 size=128
generate
1377612533:0:25:EmberEngine:Reduce:generate:37: rank=25 size=128
generate
1377622133:0:21:EmberEngine:Reduce:generate:37: rank=21 size=128
generate
1377656333:0:26:EmberEngine:Reduce:generate:37: rank=26 size=128
generate
1377665933:0:22:EmberEngine:Reduce:generate:37: rank=22 size=128
generate
1377733133:0:29:EmberEngine:Reduce:generate:37: rank=29 size=128
generate
1377776333:0:30:EmberEngine:Reduce:generate:37: rank=30 size=128
generate
1378099933:0:31:EmberEngine:Reduce:generate:37: rank=31 size=128
generate
1378230133:0:33:EmberEngine:Reduce:generate:37: rank=33 size=128
generate
1378402933:0:32:EmberEngine:Reduce:generate:37: rank=32 size=128
generate
1378446733:0:34:EmberEngine:Reduce:generate:37: rank=34 size=128
generate
1378498133:0:35:EmberEngine:Reduce:generate:37: rank=35 size=128
generate
1378507733:0:39:EmberEngine:Reduce:generate:37: rank=39 size=128
generate
1378541933:0:37:EmberEngine:Reduce:generate:37: rank=37 size=128
generate
1378551533:0:41:EmberEngine:Reduce:generate:37: rank=41 size=128
generate
1378661533:0:47:EmberEngine:Reduce:generate:37: rank=47 size=128
generate
1378705333:0:55:EmberEngine:Reduce:generate:37: rank=55 size=128
generate
1378714733:0:36:EmberEngine:Reduce:generate:37: rank=36 size=128
generate
1378733933:0:40:EmberEngine:Reduce:generate:37: rank=40 size=128
generate
1378749133:0:57:EmberEngine:Reduce:generate:37: rank=57 size=128
generate
1378757933:0:38:EmberEngine:Reduce:generate:37: rank=38 size=128
generate
1378758733:0:56:EmberEngine:Reduce:generate:37: rank=56 size=128
generate
1378767533:0:42:EmberEngine:Reduce:generate:37: rank=42 size=128
generate
1378791733:0:48:EmberEngine:Reduce:generate:37: rank=48 size=128
generate
1378792333:0:58:EmberEngine:Reduce:generate:37: rank=58 size=128
generate
1378801333:0:49:EmberEngine:Reduce:generate:37: rank=49 size=128
generate
1378819733:0:43:EmberEngine:Reduce:generate:37: rank=43 size=128
generate
1378835533:0:50:EmberEngine:Reduce:generate:37: rank=50 size=128
generate
1378835533:0:59:EmberEngine:Reduce:generate:37: rank=59 size=128
generate
1378863533:0:45:EmberEngine:Reduce:generate:37: rank=45 size=128
generate
1378878733:0:61:EmberEngine:Reduce:generate:37: rank=61 size=128
generate
1378887733:0:51:EmberEngine:Reduce:generate:37: rank=51 size=128

generate
1378888333:0:60:EmberEngine:Reduce:generate:37: rank=60 size=128
generate
1378921933:0:62:EmberEngine:Reduce:generate:37: rank=62 size=128
generate
1378931533:0:52:EmberEngine:Reduce:generate:37: rank=52 size=128
generate
1378941133:0:53:EmberEngine:Reduce:generate:37: rank=53 size=128
generate
1378974733:0:54:EmberEngine:Reduce:generate:37: rank=54 size=128
generate
1378985533:0:63:EmberEngine:Reduce:generate:37: rank=63 size=128
generate
1379026733:0:44:EmberEngine:Reduce:generate:37: rank=44 size=128
generate
1379070733:0:46:EmberEngine:Reduce:generate:37: rank=46 size=128
generate
1379072333:0:67:EmberEngine:Reduce:generate:37: rank=67 size=128
generate
1379158733:0:65:EmberEngine:Reduce:generate:37: rank=65 size=128
generate
1379168333:0:64:EmberEngine:Reduce:generate:37: rank=64 size=128
generate
1379289133:0:69:EmberEngine:Reduce:generate:37: rank=69 size=128
generate
1379298733:0:68:EmberEngine:Reduce:generate:37: rank=68 size=128
generate
1379375533:0:66:EmberEngine:Reduce:generate:37: rank=66 size=128
generate
1379426733:0:71:EmberEngine:Reduce:generate:37: rank=71 size=128
generate
1379503533:0:95:EmberEngine:Reduce:generate:37: rank=95 size=128
generate
1379505133:0:70:EmberEngine:Reduce:generate:37: rank=70 size=128
generate
1379557133:0:75:EmberEngine:Reduce:generate:37: rank=75 size=128
generate
1379609133:0:79:EmberEngine:Reduce:generate:37: rank=79 size=128
generate
1379633933:0:97:EmberEngine:Reduce:generate:37: rank=97 size=128
generate
1379643533:0:73:EmberEngine:Reduce:generate:37: rank=73 size=128
generate
1379653133:0:72:EmberEngine:Reduce:generate:37: rank=72 size=128
generate
1379653133:0:83:EmberEngine:Reduce:generate:37: rank=83 size=128
generate
1379696333:0:84:EmberEngine:Reduce:generate:37: rank=84 size=128
generate
1379729933:0:99:EmberEngine:Reduce:generate:37: rank=99 size=128
generate
1379749133:0:80:EmberEngine:Reduce:generate:37: rank=80 size=128
generate
1379764333:0:101:EmberEngine:Reduce:generate:37: rank=101 size=128
generate
1379773133:0:77:EmberEngine:Reduce:generate:37: rank=77 size=128
generate
1379782733:0:76:EmberEngine:Reduce:generate:37: rank=76 size=128
generate
1379792333:0:91:EmberEngine:Reduce:generate:37: rank=91 size=128
generate
1379806733:0:96:EmberEngine:Reduce:generate:37: rank=96 size=128
generate
1379834733:0:87:EmberEngine:Reduce:generate:37: rank=87 size=128
generate
1379835533:0:92:EmberEngine:Reduce:generate:37: rank=92 size=128
generate
1379859533:0:74:EmberEngine:Reduce:generate:37: rank=74 size=128
generate
1379860333:0:98:EmberEngine:Reduce:generate:37: rank=98 size=128
generate
1379869133:0:85:EmberEngine:Reduce:generate:37: rank=85 size=128
generate
1379878733:0:88:EmberEngine:Reduce:generate:37: rank=88 size=128
generate
1379893133:0:111:EmberEngine:Reduce:generate:37: rank=111 size=128
generate
1379893933:0:123:EmberEngine:Reduce:generate:37: rank=123 size=128
generate
1379902733:0:103:EmberEngine:Reduce:generate:37: rank=103 size=128
generate
1379912333:0:86:EmberEngine:Reduce:generate:37: rank=86 size=128

generate
1379921933:0:81:EmberEngine:Reduce:generate:37: rank=81 size=128
generate
1379937133:0:115:EmberEngine:Reduce:generate:37: rank=115 size=128
generate
1379937133:0:119:EmberEngine:Reduce:generate:37: rank=119 size=128
generate
1379937133:0:125:EmberEngine:Reduce:generate:37: rank=125 size=128
generate
1379946733:0:105:EmberEngine:Reduce:generate:37: rank=105 size=128
generate
1379946733:0:124:EmberEngine:Reduce:generate:37: rank=124 size=128
generate
1379965133:0:82:EmberEngine:Reduce:generate:37: rank=82 size=128
generate
1379965133:0:100:EmberEngine:Reduce:generate:37: rank=100 size=128
generate
1379980333:0:102:EmberEngine:Reduce:generate:37: rank=102 size=128
generate
1379980333:0:117:EmberEngine:Reduce:generate:37: rank=117 size=128
generate
1379980333:0:120:EmberEngine:Reduce:generate:37: rank=120 size=128
generate
1379989133:0:78:EmberEngine:Reduce:generate:37: rank=78 size=128
generate
1379989133:0:126:EmberEngine:Reduce:generate:37: rank=126 size=128
generate
1379989933:0:116:EmberEngine:Reduce:generate:37: rank=116 size=128
generate
1379989933:0:121:EmberEngine:Reduce:generate:37: rank=121 size=128
generate
1379999533:0:93:EmberEngine:Reduce:generate:37: rank=93 size=128
generate
1380023533:0:118:EmberEngine:Reduce:generate:37: rank=118 size=128
generate
1380033133:0:112:EmberEngine:Reduce:generate:37: rank=112 size=128
generate
1380033133:0:122:EmberEngine:Reduce:generate:37: rank=122 size=128
generate
1380041933:0:107:EmberEngine:Reduce:generate:37: rank=107 size=128
generate
1380042733:0:89:EmberEngine:Reduce:generate:37: rank=89 size=128
generate
1380042733:0:94:EmberEngine:Reduce:generate:37: rank=94 size=128
generate
1380042733:0:113:EmberEngine:Reduce:generate:37: rank=113 size=128
generate
Bcast: ranks 128, loop 2, bytes 56, latency 1.893 us
1380044333:0:127:EmberEngine:Reduce:generate:37: rank=127 size=128
generate
1380076333:0:109:EmberEngine:Reduce:generate:37: rank=109 size=128
generate
1380085933:0:90:EmberEngine:Reduce:generate:37: rank=90 size=128
generate
1380085933:0:114:EmberEngine:Reduce:generate:37: rank=114 size=128
generate
1380119533:0:104:EmberEngine:Reduce:generate:37: rank=104 size=128
generate
1380162733:0:106:EmberEngine:Reduce:generate:37: rank=106 size=128
generate
1380258733:0:108:EmberEngine:Reduce:generate:37: rank=108 size=128
generate
1380292333:0:110:EmberEngine:Reduce:generate:37: rank=110 size=128
Allgather: ranks 128, loop 3, bytes 12, latency 9.994 us
Barrier: ranks 128, loop 99, latency 14.901 us
2917697133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=1 messageSize=256 iterations=35
2918540333:0:1:EmberEngine:PingPong:generate:65: rank=1 size=128
2920705933:0:8:EmberEngine:PingPong:generate:65: rank=8 size=128
2923666733:0:64:EmberEngine:PingPong:generate:65: rank=64 size=128
2924549133:0:127:EmberEngine:PingPong:generate:65: rank=127 size=128
PingPong: otherRank 1, total time 63.549 us, loop 35, bufLen 256, latency 0.908 us. bandwidth 0.281987 GB/s
2981246733:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=8 messageSize=256 iterations=35
PingPong: otherRank 8, total time 87.416 us, loop 35, bufLen 256, latency 1.249 us. bandwidth 0.204997 GB/s
3068662733:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=64 messageSize=256 iterations=35
PingPong: otherRank 64, total time 87.416 us, loop 35, bufLen 256, latency 1.249 us. bandwidth 0.204997 GB/s
3156078733:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=127 messageSize=256 iterations=35
PingPong: otherRank 127, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918 GB/s
Simulation is complete, simulated time: 3.24504 ms

=== MiniFE torus 16x16 (PPN=2) ===

=====

Topology : Torus 16x16
Routers : 256
Nodes per router : 2 (local_ports)
Total nodes : 512
Total MPI ranks : 512
PPN (ranks/node) : 2 <-- should be 2
Stats CSV : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Torus/PPN_2/results/torus_16x16_ppn2_nodes512_stats.csv
=====

Allreduce: ranks 512, loop 92, 1 double(s), latency 23.034 us
generate
2122579133:0:0:EmberEngine:Reduce:generate:37: rank=0 size=512
generate
2123922333:0:1:EmberEngine:Reduce:generate:37: rank=1 size=512
generate
2124138333:0:2:EmberEngine:Reduce:generate:37: rank=2 size=512
generate
2124851933:0:3:EmberEngine:Reduce:generate:37: rank=3 size=512
generate
2125067933:0:5:EmberEngine:Reduce:generate:37: rank=5 size=512
generate
2125077333:0:4:EmberEngine:Reduce:generate:37: rank=4 size=512
generate
2125284533:0:6:EmberEngine:Reduce:generate:37: rank=6 size=512
generate
2125867933:0:7:EmberEngine:Reduce:generate:37: rank=7 size=512
generate
2126093333:0:8:EmberEngine:Reduce:generate:37: rank=8 size=512
generate
2126102933:0:9:EmberEngine:Reduce:generate:37: rank=9 size=512
generate
2126170333:0:11:EmberEngine:Reduce:generate:37: rank=11 size=512
generate
2126319733:0:10:EmberEngine:Reduce:generate:37: rank=10 size=512
generate
2126386933:0:13:EmberEngine:Reduce:generate:37: rank=13 size=512
generate
2126396533:0:12:EmberEngine:Reduce:generate:37: rank=12 size=512
generate
2126603533:0:14:EmberEngine:Reduce:generate:37: rank=14 size=512
generate
2127056733:0:15:EmberEngine:Reduce:generate:37: rank=15 size=512
generate
2127282133:0:16:EmberEngine:Reduce:generate:37: rank=16 size=512
generate
2127291733:0:17:EmberEngine:Reduce:generate:37: rank=17 size=512
generate
2127386933:0:19:EmberEngine:Reduce:generate:37: rank=19 size=512
generate
2127517333:0:18:EmberEngine:Reduce:generate:37: rank=18 size=512
generate
2127531933:0:23:EmberEngine:Reduce:generate:37: rank=23 size=512
generate
2127622733:0:20:EmberEngine:Reduce:generate:37: rank=20 size=512
generate
2127622933:0:21:EmberEngine:Reduce:generate:37: rank=21 size=512
generate
2127758133:0:24:EmberEngine:Reduce:generate:37: rank=24 size=512
generate
2127767733:0:25:EmberEngine:Reduce:generate:37: rank=25 size=512
generate
2127820333:0:22:EmberEngine:Reduce:generate:37: rank=22 size=512
generate
2127834933:0:27:EmberEngine:Reduce:generate:37: rank=27 size=512
generate
2127984333:0:26:EmberEngine:Reduce:generate:37: rank=26 size=512
generate
2128051533:0:29:EmberEngine:Reduce:generate:37: rank=29 size=512
generate
2128061133:0:28:EmberEngine:Reduce:generate:37: rank=28 size=512
generate
2128267533:0:30:EmberEngine:Reduce:generate:37: rank=30 size=512
generate
2128591133:0:31:EmberEngine:Reduce:generate:37: rank=31 size=512
generate
2128721333:0:32:EmberEngine:Reduce:generate:37: rank=32 size=512
generate
2128811733:0:47:EmberEngine:Reduce:generate:37: rank=47 size=512
generate
2128826133:0:35:EmberEngine:Reduce:generate:37: rank=35 size=512

generate
2128834933:0:39:EmberEngine:Reduce:generate:37: rank=39 size=512
generate
2128854933:0:48:EmberEngine:Reduce:generate:37: rank=48 size=512
generate
2128869933:0:36:EmberEngine:Reduce:generate:37: rank=36 size=512
generate
212888333:0:40:EmberEngine:Reduce:generate:37: rank=40 size=512
generate
2128902933:0:33:EmberEngine:Reduce:generate:37: rank=33 size=512
generate
2128937333:0:55:EmberEngine:Reduce:generate:37: rank=55 size=512
generate
2128946733:0:34:EmberEngine:Reduce:generate:37: rank=34 size=512
generate
2128956533:0:51:EmberEngine:Reduce:generate:37: rank=51 size=512
generate
2128981133:0:56:EmberEngine:Reduce:generate:37: rank=56 size=512
generate
2128998733:0:43:EmberEngine:Reduce:generate:37: rank=43 size=512
generate
2129000333:0:52:EmberEngine:Reduce:generate:37: rank=52 size=512
generate
2129033333:0:49:EmberEngine:Reduce:generate:37: rank=49 size=512
generate
2129042733:0:44:EmberEngine:Reduce:generate:37: rank=44 size=512
generate
2129060533:0:37:EmberEngine:Reduce:generate:37: rank=37 size=512
generate
2129067533:0:59:EmberEngine:Reduce:generate:37: rank=59 size=512
generate
2129075533:0:41:EmberEngine:Reduce:generate:37: rank=41 size=512
generate
2129077133:0:50:EmberEngine:Reduce:generate:37: rank=50 size=512
generate
2129095533:0:38:EmberEngine:Reduce:generate:37: rank=38 size=512
generate
2129110733:0:60:EmberEngine:Reduce:generate:37: rank=60 size=512
generate
2129119533:0:42:EmberEngine:Reduce:generate:37: rank=42 size=512
generate
2129163533:0:57:EmberEngine:Reduce:generate:37: rank=57 size=512
generate
2129173133:0:53:EmberEngine:Reduce:generate:37: rank=53 size=512
generate
2129181933:0:45:EmberEngine:Reduce:generate:37: rank=45 size=512
generate
2129206733:0:58:EmberEngine:Reduce:generate:37: rank=58 size=512
generate
2129216333:0:54:EmberEngine:Reduce:generate:37: rank=54 size=512
generate
2129225133:0:46:EmberEngine:Reduce:generate:37: rank=46 size=512
generate
2129283533:0:61:EmberEngine:Reduce:generate:37: rank=61 size=512
generate
2129326733:0:62:EmberEngine:Reduce:generate:37: rank=62 size=512
generate
2129520733:0:63:EmberEngine:Reduce:generate:37: rank=63 size=512
generate
2129650933:0:65:EmberEngine:Reduce:generate:37: rank=65 size=512
generate
2129823733:0:64:EmberEngine:Reduce:generate:37: rank=64 size=512
generate
2129867533:0:66:EmberEngine:Reduce:generate:37: rank=66 size=512
generate
2129918933:0:67:EmberEngine:Reduce:generate:37: rank=67 size=512
generate
2129928533:0:71:EmberEngine:Reduce:generate:37: rank=71 size=512
generate
2129962733:0:69:EmberEngine:Reduce:generate:37: rank=69 size=512
generate
2129972333:0:73:EmberEngine:Reduce:generate:37: rank=73 size=512
generate
2130110733:0:79:EmberEngine:Reduce:generate:37: rank=79 size=512
generate
2130135533:0:68:EmberEngine:Reduce:generate:37: rank=68 size=512
generate
2130145133:0:72:EmberEngine:Reduce:generate:37: rank=72 size=512
generate
2130163533:0:81:EmberEngine:Reduce:generate:37: rank=81 size=512
generate
2130178733:0:70:EmberEngine:Reduce:generate:37: rank=70 size=512

generate
2130188333:0:74:EmberEngine:Reduce:generate:37: rank=74 size=512
generate
2130249333:0:75:EmberEngine:Reduce:generate:37: rank=75 size=512
generate
2130256333:0:119:EmberEngine:Reduce:generate:37: rank=119 size=512
generate
2130284333:0:77:EmberEngine:Reduce:generate:37: rank=77 size=512
generate
2130299533:0:121:EmberEngine:Reduce:generate:37: rank=121 size=512
generate
2130303533:0:111:EmberEngine:Reduce:generate:37: rank=111 size=512
generate
2130309133:0:120:EmberEngine:Reduce:generate:37: rank=120 size=512
generate
2130327533:0:80:EmberEngine:Reduce:generate:37: rank=80 size=512
generate
2130342733:0:113:EmberEngine:Reduce:generate:37: rank=113 size=512
generate
2130342733:0:122:EmberEngine:Reduce:generate:37: rank=122 size=512
generate
2130352333:0:112:EmberEngine:Reduce:generate:37: rank=112 size=512
generate
2130380333:0:82:EmberEngine:Reduce:generate:37: rank=82 size=512
generate
2130385933:0:114:EmberEngine:Reduce:generate:37: rank=114 size=512
generate
2130385933:0:123:EmberEngine:Reduce:generate:37: rank=123 size=512
generate
2130429133:0:125:EmberEngine:Reduce:generate:37: rank=125 size=512
generate
2130432533:0:95:EmberEngine:Reduce:generate:37: rank=95 size=512
generate
2130437133:0:83:EmberEngine:Reduce:generate:37: rank=83 size=512
generate
2130438733:0:115:EmberEngine:Reduce:generate:37: rank=115 size=512
generate
2130438733:0:124:EmberEngine:Reduce:generate:37: rank=124 size=512
generate
2130448333:0:87:EmberEngine:Reduce:generate:37: rank=87 size=512
generate
2130465933:0:76:EmberEngine:Reduce:generate:37: rank=76 size=512
generate
2130472333:0:126:EmberEngine:Reduce:generate:37: rank=126 size=512
generate
2130481133:0:85:EmberEngine:Reduce:generate:37: rank=85 size=512
generate
2130481933:0:116:EmberEngine:Reduce:generate:37: rank=116 size=512
generate
2130490733:0:89:EmberEngine:Reduce:generate:37: rank=89 size=512
generate
2130490933:0:103:EmberEngine:Reduce:generate:37: rank=103 size=512
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2130491533:0:117:EmberEngine:Reduce:generate:37: rank=117 size=512
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2130500333:0:78:EmberEngine:Reduce:generate:37: rank=78 size=512
generate
2130525133:0:118:EmberEngine:Reduce:generate:37: rank=118 size=512
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2130536733:0:127:EmberEngine:Reduce:generate:37: rank=127 size=512
generate
2130544333:0:105:EmberEngine:Reduce:generate:37: rank=105 size=512
generate
2130553933:0:104:EmberEngine:Reduce:generate:37: rank=104 size=512
generate
2130571533:0:96:EmberEngine:Reduce:generate:37: rank=96 size=512
generate
2130571733:0:97:EmberEngine:Reduce:generate:37: rank=97 size=512
generate
2130577933:0:106:EmberEngine:Reduce:generate:37: rank=106 size=512
generate
2130605933:0:98:EmberEngine:Reduce:generate:37: rank=98 size=512
generate
2130621133:0:107:EmberEngine:Reduce:generate:37: rank=107 size=512
generate
2130653133:0:84:EmberEngine:Reduce:generate:37: rank=84 size=512
generate
2130654133:0:99:EmberEngine:Reduce:generate:37: rank=99 size=512
generate
2130663533:0:88:EmberEngine:Reduce:generate:37: rank=88 size=512
generate
2130664333:0:108:EmberEngine:Reduce:generate:37: rank=108 size=512

generate
2130673933:0:109:EmberEngine:Reduce:generate:37: rank=109 size=512
generate
2130697133:0:86:EmberEngine:Reduce:generate:37: rank=86 size=512
generate
2130697933:0:100:EmberEngine:Reduce:generate:37: rank=100 size=512
generate
2130706733:0:90:EmberEngine:Reduce:generate:37: rank=90 size=512
generate
2130707533:0:101:EmberEngine:Reduce:generate:37: rank=101 size=512
generate
2130707533:0:110:EmberEngine:Reduce:generate:37: rank=110 size=512
generate
2130724533:0:91:EmberEngine:Reduce:generate:37: rank=91 size=512
generate
2130741133:0:102:EmberEngine:Reduce:generate:37: rank=102 size=512
generate
2130753333:0:131:EmberEngine:Reduce:generate:37: rank=131 size=512
generate
2130759533:0:93:EmberEngine:Reduce:generate:37: rank=93 size=512
generate
2130839733:0:129:EmberEngine:Reduce:generate:37: rank=129 size=512
generate
2130849333:0:128:EmberEngine:Reduce:generate:37: rank=128 size=512
generate
2130932333:0:92:EmberEngine:Reduce:generate:37: rank=92 size=512
generate
2130969933:0:133:EmberEngine:Reduce:generate:37: rank=133 size=512
generate
2130975533:0:94:EmberEngine:Reduce:generate:37: rank=94 size=512
generate
2130979533:0:132:EmberEngine:Reduce:generate:37: rank=132 size=512
generate
2131056333:0:130:EmberEngine:Reduce:generate:37: rank=130 size=512
generate
2131107733:0:135:EmberEngine:Reduce:generate:37: rank=135 size=512
generate
2131185933:0:134:EmberEngine:Reduce:generate:37: rank=134 size=512
generate
2131237933:0:139:EmberEngine:Reduce:generate:37: rank=139 size=512
generate
2131290133:0:143:EmberEngine:Reduce:generate:37: rank=143 size=512
generate
2131324333:0:137:EmberEngine:Reduce:generate:37: rank=137 size=512
generate
2131333933:0:136:EmberEngine:Reduce:generate:37: rank=136 size=512
generate
2131420333:0:147:EmberEngine:Reduce:generate:37: rank=147 size=512
generate
2131453933:0:141:EmberEngine:Reduce:generate:37: rank=141 size=512
generate
2131473133:0:140:EmberEngine:Reduce:generate:37: rank=140 size=512
generate
2131506733:0:145:EmberEngine:Reduce:generate:37: rank=145 size=512
generate
2131516333:0:144:EmberEngine:Reduce:generate:37: rank=144 size=512
generate
2131536333:0:191:EmberEngine:Reduce:generate:37: rank=191 size=512
generate
2131540333:0:138:EmberEngine:Reduce:generate:37: rank=138 size=512
generate
2131595133:0:255:EmberEngine:Reduce:generate:37: rank=255 size=512
generate
2131636333:0:149:EmberEngine:Reduce:generate:37: rank=149 size=512
generate
2131655533:0:148:EmberEngine:Reduce:generate:37: rank=148 size=512
generate
2131669933:0:142:EmberEngine:Reduce:generate:37: rank=142 size=512
generate
2131689933:0:193:EmberEngine:Reduce:generate:37: rank=193 size=512
generate
2131704333:0:251:EmberEngine:Reduce:generate:37: rank=251 size=512
generate
2131722733:0:146:EmberEngine:Reduce:generate:37: rank=146 size=512
generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
2132987533:0:283:EmberEngine:Reduce:generate:37: rank=283 size=512
generate
Bcast: ranks 512, loop 2, bytes 56, latency 1.893 us
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate

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generate
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generate
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generate
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generate
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generate
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generate

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generate
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2133586733:0:332:EmberEngine:Reduce:generate:37: rank=332 size=512
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generate
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generate

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generate
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generate
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generate
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generate
2133614733:0:400:EmberEngine:Reduce:generate:37: rank=400 size=512
generate
2133621133:0:334:EmberEngine:Reduce:generate:37: rank=334 size=512
generate
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generate
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generate
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generate
2133649133:0:336:EmberEngine:Reduce:generate:37: rank=336 size=512
generate
2133651533:0:490:EmberEngine:Reduce:generate:37: rank=490 size=512
generate
2133653933:0:374:EmberEngine:Reduce:generate:37: rank=374 size=512
generate
2133663533:0:365:EmberEngine:Reduce:generate:37: rank=365 size=512
generate
2133669933:0:479:EmberEngine:Reduce:generate:37: rank=479 size=512
generate
2133673133:0:461:EmberEngine:Reduce:generate:37: rank=461 size=512
generate
2133679533:0:463:EmberEngine:Reduce:generate:37: rank=463 size=512
generate
2133688333:0:366:EmberEngine:Reduce:generate:37: rank=366 size=512
generate
2133689133:0:407:EmberEngine:Reduce:generate:37: rank=407 size=512
generate
2133689133:0:443:EmberEngine:Reduce:generate:37: rank=443 size=512
generate
2133692333:0:338:EmberEngine:Reduce:generate:37: rank=338 size=512
generate
2133713133:0:456:EmberEngine:Reduce:generate:37: rank=456 size=512
generate
2133722733:0:465:EmberEngine:Reduce:generate:37: rank=465 size=512
generate
2133725933:0:339:EmberEngine:Reduce:generate:37: rank=339 size=512
generate
2133726733:0:405:EmberEngine:Reduce:generate:37: rank=405 size=512
generate
2133731533:0:415:EmberEngine:Reduce:generate:37: rank=415 size=512
generate
2133732333:0:435:EmberEngine:Reduce:generate:37: rank=435 size=512
generate
2133732333:0:444:EmberEngine:Reduce:generate:37: rank=444 size=512
generate
2133733133:0:452:EmberEngine:Reduce:generate:37: rank=452 size=512
generate
2133740333:0:352:EmberEngine:Reduce:generate:37: rank=352 size=512
generate
2133741933:0:427:EmberEngine:Reduce:generate:37: rank=427 size=512
generate
2133749933:0:353:EmberEngine:Reduce:generate:37: rank=353 size=512
generate
2133759533:0:404:EmberEngine:Reduce:generate:37: rank=404 size=512
generate
2133760333:0:439:EmberEngine:Reduce:generate:37: rank=439 size=512
generate
2133760333:0:483:EmberEngine:Reduce:generate:37: rank=483 size=512
generate
2133765933:0:458:EmberEngine:Reduce:generate:37: rank=458 size=512
generate
2133769133:0:341:EmberEngine:Reduce:generate:37: rank=341 size=512
generate
2133775533:0:431:EmberEngine:Reduce:generate:37: rank=431 size=512
generate
2133775533:0:436:EmberEngine:Reduce:generate:37: rank=436 size=512
generate
2133775533:0:454:EmberEngine:Reduce:generate:37: rank=454 size=512
generate

2133782733:0:344:EmberEngine:Reduce:generate:37: rank=344 size=512
generate
2133785133:0:423:EmberEngine:Reduce:generate:37: rank=423 size=512
generate
2133785133:0:428:EmberEngine:Reduce:generate:37: rank=428 size=512
generate
2133793133:0:354:EmberEngine:Reduce:generate:37: rank=354 size=512
generate
2133794733:0:419:EmberEngine:Reduce:generate:37: rank=419 size=512
generate
2133799533:0:480:EmberEngine:Reduce:generate:37: rank=480 size=512
generate
2133800333:0:485:EmberEngine:Reduce:generate:37: rank=485 size=512
generate
2133802733:0:440:EmberEngine:Reduce:generate:37: rank=440 size=512
generate
2133809133:0:402:EmberEngine:Reduce:generate:37: rank=402 size=512
generate
2133809133:0:411:EmberEngine:Reduce:generate:37: rank=411 size=512
generate
2133809933:0:484:EmberEngine:Reduce:generate:37: rank=484 size=512
generate
2133817933:0:346:EmberEngine:Reduce:generate:37: rank=346 size=512
generate
2133817933:0:355:EmberEngine:Reduce:generate:37: rank=355 size=512
generate
2133828333:0:432:EmberEngine:Reduce:generate:37: rank=432 size=512
generate
2133828333:0:467:EmberEngine:Reduce:generate:37: rank=467 size=512
generate
2133828333:0:481:EmberEngine:Reduce:generate:37: rank=481 size=512
generate
2133837133:0:471:EmberEngine:Reduce:generate:37: rank=471 size=512
generate
2133837133:0:486:EmberEngine:Reduce:generate:37: rank=486 size=512
generate
2133837933:0:420:EmberEngine:Reduce:generate:37: rank=420 size=512
generate
2133837933:0:424:EmberEngine:Reduce:generate:37: rank=424 size=512
generate
2133842733:0:460:EmberEngine:Reduce:generate:37: rank=460 size=512
generate
2133852333:0:469:EmberEngine:Reduce:generate:37: rank=469 size=512
generate
2133859533:0:347:EmberEngine:Reduce:generate:37: rank=347 size=512
generate
2133861133:0:356:EmberEngine:Reduce:generate:37: rank=356 size=512
generate
2133861933:0:416:EmberEngine:Reduce:generate:37: rank=416 size=512
generate
2133871533:0:473:EmberEngine:Reduce:generate:37: rank=473 size=512
generate
2133877133:0:482:EmberEngine:Reduce:generate:37: rank=482 size=512
generate
2133880333:0:357:EmberEngine:Reduce:generate:37: rank=357 size=512
generate
2133889933:0:462:EmberEngine:Reduce:generate:37: rank=462 size=512
generate
2133894733:0:349:EmberEngine:Reduce:generate:37: rank=349 size=512
generate
2133895533:0:464:EmberEngine:Reduce:generate:37: rank=464 size=512
generate
2133905133:0:445:EmberEngine:Reduce:generate:37: rank=445 size=512
generate
2133909133:0:409:EmberEngine:Reduce:generate:37: rank=409 size=512
generate
2133913133:0:358:EmberEngine:Reduce:generate:37: rank=358 size=512
generate
2133918733:0:408:EmberEngine:Reduce:generate:37: rank=408 size=512
generate
2133937133:0:340:EmberEngine:Reduce:generate:37: rank=340 size=512
generate
2133941933:0:466:EmberEngine:Reduce:generate:37: rank=466 size=512
generate
2133942733:0:406:EmberEngine:Reduce:generate:37: rank=406 size=512
generate
2133948333:0:446:EmberEngine:Reduce:generate:37: rank=446 size=512
generate
2133948333:0:475:EmberEngine:Reduce:generate:37: rank=475 size=512
generate
2133961133:0:437:EmberEngine:Reduce:generate:37: rank=437 size=512
generate

```

2133967533:0:429:EmberEngine:Reduce:generate:37: rank=429 size=512
generate
2133976333:0:441:EmberEngine:Reduce:generate:37: rank=441 size=512
generate
2133981133:0:342:EmberEngine:Reduce:generate:37: rank=342 size=512
generate
2133981933:0:477:EmberEngine:Reduce:generate:37: rank=477 size=512
generate
2133991533:0:433:EmberEngine:Reduce:generate:37: rank=433 size=512
generate
2133991533:0:438:EmberEngine:Reduce:generate:37: rank=438 size=512
generate
2134001133:0:425:EmberEngine:Reduce:generate:37: rank=425 size=512
generate
2134001133:0:430:EmberEngine:Reduce:generate:37: rank=430 size=512
generate
2134010733:0:442:EmberEngine:Reduce:generate:37: rank=442 size=512
generate
2134020333:0:421:EmberEngine:Reduce:generate:37: rank=421 size=512
generate
2134025133:0:413:EmberEngine:Reduce:generate:37: rank=413 size=512
generate
2134034733:0:412:EmberEngine:Reduce:generate:37: rank=412 size=512
generate
2134034733:0:417:EmberEngine:Reduce:generate:37: rank=417 size=512
generate
2134034733:0:434:EmberEngine:Reduce:generate:37: rank=434 size=512
generate
2134037933:0:468:EmberEngine:Reduce:generate:37: rank=468 size=512
generate
2134044333:0:426:EmberEngine:Reduce:generate:37: rank=426 size=512
generate
2134053133:0:472:EmberEngine:Reduce:generate:37: rank=472 size=512
generate
2134053933:0:422:EmberEngine:Reduce:generate:37: rank=422 size=512
generate
2134068333:0:470:EmberEngine:Reduce:generate:37: rank=470 size=512
generate
2134075533:0:348:EmberEngine:Reduce:generate:37: rank=348 size=512
generate
2134077933:0:418:EmberEngine:Reduce:generate:37: rank=418 size=512
generate
2134087533:0:474:EmberEngine:Reduce:generate:37: rank=474 size=512
generate
2134110733:0:350:EmberEngine:Reduce:generate:37: rank=350 size=512
generate
2134125133:0:410:EmberEngine:Reduce:generate:37: rank=410 size=512
generate
2134164333:0:476:EmberEngine:Reduce:generate:37: rank=476 size=512
generate
2134197933:0:478:EmberEngine:Reduce:generate:37: rank=478 size=512
generate
2134241133:0:414:EmberEngine:Reduce:generate:37: rank=414 size=512
Allgather: ranks 512, loop 3, bytes 12, latency 15.510 us
Barrier: ranks 512, loop 99, latency 23.044 us
4499078733:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=1 messageSize=256 iterations=35
4499921933:0:1:EmberEngine:PingPong:generate:65: rank=1 size=512
4503276333:0:16:EmberEngine:PingPong:generate:65: rank=16 size=512
4507767533:0:256:EmberEngine:PingPong:generate:65: rank=256 size=512
4508999533:0:511:EmberEngine:PingPong:generate:65: rank=511 size=512
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.282320 GB/s
4562552333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=16 messageSize=256 iterations=35
PingPong: otherRank 16, total time 111.608 us, loop 35, bufLen 256, latency 1.594 us. bandwidth 0.160562 GB/s
4674160333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=256 messageSize=256 iterations=35
PingPong: otherRank 256, total time 111.608 us, loop 35, bufLen 256, latency 1.594 us. bandwidth 0.160562 GB/s
4785768333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=511 messageSize=256 iterations=35
PingPong: otherRank 511, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918 GB/s
Simulation is complete, simulated time: 4.88186 ms
=== MiniFE torus 32x32 (PPN=2) ===
=====
Topology          : Torus 32x32
Routers           : 1024
Nodes per router  : 2 (local_ports)
Total nodes       : 2048
Total MPI ranks   : 2048
PPN (ranks/node) : 2 <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Torus/PPN_2/results/torus_
32x32_ppn2_nodes2048_stats.csv

```

=====

Allreduce: ranks 2048, loop 92, 1 double(s), latency 36.429 us
generate
3354914333:0:0:EmberEngine:Reduce:generate:37: rank=0 size=2048
generate
3356257533:0:1:EmberEngine:Reduce:generate:37: rank=1 size=2048
generate
3356473533:0:2:EmberEngine:Reduce:generate:37: rank=2 size=2048
generate
3357187133:0:3:EmberEngine:Reduce:generate:37: rank=3 size=2048
generate
3357403133:0:5:EmberEngine:Reduce:generate:37: rank=5 size=2048
generate
3357412533:0:4:EmberEngine:Reduce:generate:37: rank=4 size=2048
generate
3357619733:0:6:EmberEngine:Reduce:generate:37: rank=6 size=2048
generate
3358203133:0:7:EmberEngine:Reduce:generate:37: rank=7 size=2048
generate
3358428533:0:8:EmberEngine:Reduce:generate:37: rank=8 size=2048
generate
3358438133:0:9:EmberEngine:Reduce:generate:37: rank=9 size=2048
generate
3358505533:0:11:EmberEngine:Reduce:generate:37: rank=11 size=2048
generate
3358654933:0:10:EmberEngine:Reduce:generate:37: rank=10 size=2048
generate
3358722133:0:13:EmberEngine:Reduce:generate:37: rank=13 size=2048
generate
3358731733:0:12:EmberEngine:Reduce:generate:37: rank=12 size=2048
generate
3358938733:0:14:EmberEngine:Reduce:generate:37: rank=14 size=2048
generate
3359391933:0:15:EmberEngine:Reduce:generate:37: rank=15 size=2048
generate
3359617333:0:16:EmberEngine:Reduce:generate:37: rank=16 size=2048
generate
3359626933:0:17:EmberEngine:Reduce:generate:37: rank=17 size=2048
generate
3359722133:0:19:EmberEngine:Reduce:generate:37: rank=19 size=2048
generate
3359852533:0:18:EmberEngine:Reduce:generate:37: rank=18 size=2048
generate
3359867133:0:23:EmberEngine:Reduce:generate:37: rank=23 size=2048
generate
3359957933:0:20:EmberEngine:Reduce:generate:37: rank=20 size=2048
generate
3359958133:0:21:EmberEngine:Reduce:generate:37: rank=21 size=2048
generate
3360093333:0:24:EmberEngine:Reduce:generate:37: rank=24 size=2048
generate
3360102933:0:25:EmberEngine:Reduce:generate:37: rank=25 size=2048
generate
3360155533:0:22:EmberEngine:Reduce:generate:37: rank=22 size=2048
generate
3360170133:0:27:EmberEngine:Reduce:generate:37: rank=27 size=2048
generate
3360319533:0:26:EmberEngine:Reduce:generate:37: rank=26 size=2048
generate
3360386733:0:29:EmberEngine:Reduce:generate:37: rank=29 size=2048
generate
3360396333:0:28:EmberEngine:Reduce:generate:37: rank=28 size=2048
generate
3360602733:0:30:EmberEngine:Reduce:generate:37: rank=30 size=2048
generate
3360926333:0:31:EmberEngine:Reduce:generate:37: rank=31 size=2048
generate
3361151733:0:32:EmberEngine:Reduce:generate:37: rank=32 size=2048
generate
3361161333:0:33:EmberEngine:Reduce:generate:37: rank=33 size=2048
generate
3361256533:0:35:EmberEngine:Reduce:generate:37: rank=35 size=2048
generate
3361386933:0:34:EmberEngine:Reduce:generate:37: rank=34 size=2048
generate
3361438933:0:39:EmberEngine:Reduce:generate:37: rank=39 size=2048
generate
3361492533:0:36:EmberEngine:Reduce:generate:37: rank=36 size=2048
generate
3361510933:0:37:EmberEngine:Reduce:generate:37: rank=37 size=2048
generate
3361683533:0:40:EmberEngine:Reduce:generate:37: rank=40 size=2048

generate
3361693333:0:41:EmberEngine:Reduce:generate:37: rank=41 size=2048
generate
3361727533:0:38:EmberEngine:Reduce:generate:37: rank=38 size=2048
generate
3361747133:0:47:EmberEngine:Reduce:generate:37: rank=47 size=2048
generate
3361789333:0:43:EmberEngine:Reduce:generate:37: rank=43 size=2048
generate
3361909933:0:42:EmberEngine:Reduce:generate:37: rank=42 size=2048
generate
3361973333:0:48:EmberEngine:Reduce:generate:37: rank=48 size=2048
generate
3361977133:0:45:EmberEngine:Reduce:generate:37: rank=45 size=2048
generate
3361982933:0:49:EmberEngine:Reduce:generate:37: rank=49 size=2048
generate
3362005933:0:44:EmberEngine:Reduce:generate:37: rank=44 size=2048
generate
3362087733:0:51:EmberEngine:Reduce:generate:37: rank=51 size=2048
generate
3362174733:0:46:EmberEngine:Reduce:generate:37: rank=46 size=2048
generate
3362208333:0:50:EmberEngine:Reduce:generate:37: rank=50 size=2048
generate
3362222933:0:55:EmberEngine:Reduce:generate:37: rank=55 size=2048
generate
3362304333:0:53:EmberEngine:Reduce:generate:37: rank=53 size=2048
generate
3362313933:0:52:EmberEngine:Reduce:generate:37: rank=52 size=2048
generate
3362449133:0:56:EmberEngine:Reduce:generate:37: rank=56 size=2048
generate
3362458733:0:57:EmberEngine:Reduce:generate:37: rank=57 size=2048
generate
3362510733:0:54:EmberEngine:Reduce:generate:37: rank=54 size=2048
generate
3362525933:0:59:EmberEngine:Reduce:generate:37: rank=59 size=2048
generate
3362674733:0:58:EmberEngine:Reduce:generate:37: rank=58 size=2048
generate
3362741933:0:61:EmberEngine:Reduce:generate:37: rank=61 size=2048
generate
3362751533:0:60:EmberEngine:Reduce:generate:37: rank=60 size=2048
generate
3362957933:0:62:EmberEngine:Reduce:generate:37: rank=62 size=2048
generate
3363151933:0:63:EmberEngine:Reduce:generate:37: rank=63 size=2048
generate
3363282133:0:64:EmberEngine:Reduce:generate:37: rank=64 size=2048
generate
3363376533:0:95:EmberEngine:Reduce:generate:37: rank=95 size=2048
generate
3363386933:0:67:EmberEngine:Reduce:generate:37: rank=67 size=2048
generate
3363395733:0:71:EmberEngine:Reduce:generate:37: rank=71 size=2048
generate
3363405333:0:79:EmberEngine:Reduce:generate:37: rank=79 size=2048
generate
3363416733:0:96:EmberEngine:Reduce:generate:37: rank=96 size=2048
generate
3363430733:0:68:EmberEngine:Reduce:generate:37: rank=68 size=2048
generate
3363449133:0:72:EmberEngine:Reduce:generate:37: rank=72 size=2048
generate
3363449133:0:80:EmberEngine:Reduce:generate:37: rank=80 size=2048
generate
3363463733:0:65:EmberEngine:Reduce:generate:37: rank=65 size=2048
generate
3363502733:0:111:EmberEngine:Reduce:generate:37: rank=111 size=2048
generate
3363507533:0:66:EmberEngine:Reduce:generate:37: rank=66 size=2048
generate
3363517333:0:99:EmberEngine:Reduce:generate:37: rank=99 size=2048
generate
3363535733:0:103:EmberEngine:Reduce:generate:37: rank=103 size=2048
generate
3363545933:0:112:EmberEngine:Reduce:generate:37: rank=112 size=2048
generate
3363561133:0:100:EmberEngine:Reduce:generate:37: rank=100 size=2048
generate
3363563733:0:75:EmberEngine:Reduce:generate:37: rank=75 size=2048

generate
3363573333:0:83:EmberEngine:Reduce:generate:37: rank=83 size=2048
generate
3363579533:0:104:EmberEngine:Reduce:generate:37: rank=104 size=2048
generate
3363582933:0:87:EmberEngine:Reduce:generate:37: rank=87 size=2048
generate
3363594133:0:97:EmberEngine:Reduce:generate:37: rank=97 size=2048
generate
3363607533:0:76:EmberEngine:Reduce:generate:37: rank=76 size=2048
generate
3363616533:0:84:EmberEngine:Reduce:generate:37: rank=84 size=2048
generate
3363621333:0:69:EmberEngine:Reduce:generate:37: rank=69 size=2048
generate
3363628333:0:119:EmberEngine:Reduce:generate:37: rank=119 size=2048
generate
3363631733:0:73:EmberEngine:Reduce:generate:37: rank=73 size=2048
generate
3363636333:0:88:EmberEngine:Reduce:generate:37: rank=88 size=2048
generate
3363637933:0:98:EmberEngine:Reduce:generate:37: rank=98 size=2048
generate
3363647533:0:115:EmberEngine:Reduce:generate:37: rank=115 size=2048
generate
3363649933:0:81:EmberEngine:Reduce:generate:37: rank=81 size=2048
generate
3363656333:0:70:EmberEngine:Reduce:generate:37: rank=70 size=2048
generate
3363665933:0:107:EmberEngine:Reduce:generate:37: rank=107 size=2048
generate
3363671533:0:120:EmberEngine:Reduce:generate:37: rank=120 size=2048
generate
3363675533:0:74:EmberEngine:Reduce:generate:37: rank=74 size=2048
generate
3363684333:0:91:EmberEngine:Reduce:generate:37: rank=91 size=2048
generate
3363690733:0:116:EmberEngine:Reduce:generate:37: rank=116 size=2048
generate
3363693133:0:82:EmberEngine:Reduce:generate:37: rank=82 size=2048
generate
3363709133:0:108:EmberEngine:Reduce:generate:37: rank=108 size=2048
generate
3363724333:0:113:EmberEngine:Reduce:generate:37: rank=113 size=2048
generate
3363727533:0:92:EmberEngine:Reduce:generate:37: rank=92 size=2048
generate
3363742733:0:101:EmberEngine:Reduce:generate:37: rank=101 size=2048
generate
3363757933:0:123:EmberEngine:Reduce:generate:37: rank=123 size=2048
generate
3363767533:0:114:EmberEngine:Reduce:generate:37: rank=114 size=2048
generate
3363771533:0:105:EmberEngine:Reduce:generate:37: rank=105 size=2048
generate
3363780333:0:77:EmberEngine:Reduce:generate:37: rank=77 size=2048
generate
3363785933:0:102:EmberEngine:Reduce:generate:37: rank=102 size=2048
generate
3363789933:0:85:EmberEngine:Reduce:generate:37: rank=85 size=2048
generate
3363799533:0:89:EmberEngine:Reduce:generate:37: rank=89 size=2048
generate
3363801133:0:124:EmberEngine:Reduce:generate:37: rank=124 size=2048
generate
3363814733:0:106:EmberEngine:Reduce:generate:37: rank=106 size=2048
generate
3363823533:0:78:EmberEngine:Reduce:generate:37: rank=78 size=2048
generate
3363833133:0:86:EmberEngine:Reduce:generate:37: rank=86 size=2048
generate
3363842733:0:90:EmberEngine:Reduce:generate:37: rank=90 size=2048
generate
3363853933:0:121:EmberEngine:Reduce:generate:37: rank=121 size=2048
generate
3363863533:0:117:EmberEngine:Reduce:generate:37: rank=117 size=2048
generate
3363872333:0:109:EmberEngine:Reduce:generate:37: rank=109 size=2048
generate
3363881933:0:93:EmberEngine:Reduce:generate:37: rank=93 size=2048
generate
3363897133:0:122:EmberEngine:Reduce:generate:37: rank=122 size=2048

generate
3363906733:0:118:EmberEngine:Reduce:generate:37: rank=118 size=2048
generate
3363915533:0:110:EmberEngine:Reduce:generate:37: rank=110 size=2048
generate
3363925133:0:94:EmberEngine:Reduce:generate:37: rank=94 size=2048
generate
3363973933:0:125:EmberEngine:Reduce:generate:37: rank=125 size=2048
generate
3364017133:0:126:EmberEngine:Reduce:generate:37: rank=126 size=2048
generate
3364081533:0:127:EmberEngine:Reduce:generate:37: rank=127 size=2048
generate
3364211733:0:129:EmberEngine:Reduce:generate:37: rank=129 size=2048
generate
3364384533:0:128:EmberEngine:Reduce:generate:37: rank=128 size=2048
generate
3364428333:0:130:EmberEngine:Reduce:generate:37: rank=130 size=2048
generate
3364479733:0:131:EmberEngine:Reduce:generate:37: rank=131 size=2048
generate
3364489333:0:135:EmberEngine:Reduce:generate:37: rank=135 size=2048
generate
3364523533:0:133:EmberEngine:Reduce:generate:37: rank=133 size=2048
generate
3364533133:0:137:EmberEngine:Reduce:generate:37: rank=137 size=2048
generate
3364675733:0:143:EmberEngine:Reduce:generate:37: rank=143 size=2048
generate
3364696333:0:132:EmberEngine:Reduce:generate:37: rank=132 size=2048
generate
3364705933:0:136:EmberEngine:Reduce:generate:37: rank=136 size=2048
generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
3369776333:0:1535:EmberEngine:Reduce:generate:37: rank=1535 size=2048
generate
3369782733:0:888:EmberEngine:Reduce:generate:37: rank=888 size=2048
generate
Bcast: ranks 2048, loop 2, bytes 56, latency 1.893 us
3369785933:0:2047:EmberEngine:Reduce:generate:37: rank=2047 size=2048
generate
3369789133:0:675:EmberEngine:Reduce:generate:37: rank=675 size=2048
generate
3369793933:0:871:EmberEngine:Reduce:generate:37: rank=871 size=2048
generate
3369793933:0:876:EmberEngine:Reduce:generate:37: rank=876 size=2048
generate
3369794733:0:867:EmberEngine:Reduce:generate:37: rank=867 size=2048
generate
3369794733:0:963:EmberEngine:Reduce:generate:37: rank=963 size=2048
generate
3369794933:0:808:EmberEngine:Reduce:generate:37: rank=808 size=2048
generate
3369806133:0:914:EmberEngine:Reduce:generate:37: rank=914 size=2048
generate
3369809133:0:731:EmberEngine:Reduce:generate:37: rank=731 size=2048
generate
3369810133:0:923:EmberEngine:Reduce:generate:37: rank=923 size=2048
generate
3369813933:0:809:EmberEngine:Reduce:generate:37: rank=809 size=2048
generate
3369817133:0:660:EmberEngine:Reduce:generate:37: rank=660 size=2048
generate
3369823533:0:1071:EmberEngine:Reduce:generate:37: rank=1071 size=2048
generate
3369824333:0:930:EmberEngine:Reduce:generate:37: rank=930 size=2048
generate

3369824333:0:939:EmberEngine:Reduce:generate:37: rank=939 size=2048
generate
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generate
3369827533:0:868:EmberEngine:Reduce:generate:37: rank=868 size=2048
generate
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generate
3369828533:0:863:EmberEngine:Reduce:generate:37: rank=863 size=2048
generate
3369830133:0:880:EmberEngine:Reduce:generate:37: rank=880 size=2048
generate
3369837933:0:677:EmberEngine:Reduce:generate:37: rank=677 size=2048
generate
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generate
3369846733:0:964:EmberEngine:Reduce:generate:37: rank=964 size=2048
generate
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generate
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generate
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generate
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generate
3369858733:0:925:EmberEngine:Reduce:generate:37: rank=925 size=2048
generate
3369861933:0:960:EmberEngine:Reduce:generate:37: rank=960 size=2048
generate
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generate
3369868333:0:966:EmberEngine:Reduce:generate:37: rank=966 size=2048
generate
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generate
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generate

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generate

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generate
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generate

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generate
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generate
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generate
3372557933:0:1650:EmberEngine:Reduce:generate:37: rank=1650 size=2048
generate
3372558733:0:1832:EmberEngine:Reduce:generate:37: rank=1832 size=2048
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3372561133:0:1604:EmberEngine:Reduce:generate:37: rank=1604 size=2048
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3372561133:0:1901:EmberEngine:Reduce:generate:37: rank=1901 size=2048
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3372561933:0:1386:EmberEngine:Reduce:generate:37: rank=1386 size=2048
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3372563533:0:1884:EmberEngine:Reduce:generate:37: rank=1884 size=2048
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3372571533:0:1453:EmberEngine:Reduce:generate:37: rank=1453 size=2048
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3372584333:0:1618:EmberEngine:Reduce:generate:37: rank=1618 size=2048
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3372590733:0:1370:EmberEngine:Reduce:generate:37: rank=1370 size=2048
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3372590733:0:1460:EmberEngine:Reduce:generate:37: rank=1460 size=2048
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3372592333:0:1980:EmberEngine:Reduce:generate:37: rank=1980 size=2048
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3372596333:0:1756:EmberEngine:Reduce:generate:37: rank=1756 size=2048
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3372619533:0:1893:EmberEngine:Reduce:generate:37: rank=1893 size=2048
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3372620333:0:1758:EmberEngine:Reduce:generate:37: rank=1758 size=2048
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3372620333:0:1876:EmberEngine:Reduce:generate:37: rank=1876 size=2048
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3372623533:0:1743:EmberEngine:Reduce:generate:37: rank=1743 size=2048
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3372625933:0:1982:EmberEngine:Reduce:generate:37: rank=1982 size=2048
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3372628333:0:1448:EmberEngine:Reduce:generate:37: rank=1448 size=2048
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3372628333:0:1602:EmberEngine:Reduce:generate:37: rank=1602 size=2048
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3372635533:0:1653:EmberEngine:Reduce:generate:37: rank=1653 size=2048
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3372652333:0:1346:EmberEngine:Reduce:generate:37: rank=1346 size=2048
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3372653133:0:1690:EmberEngine:Reduce:generate:37: rank=1690 size=2048
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3372653933:0:1843:EmberEngine:Reduce:generate:37: rank=1843 size=2048
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3372653933:0:1894:EmberEngine:Reduce:generate:37: rank=1894 size=2048
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3372655533:0:1646:EmberEngine:Reduce:generate:37: rank=1646 size=2048
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3372665933:0:1450:EmberEngine:Reduce:generate:37: rank=1450 size=2048
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3372667533:0:1621:EmberEngine:Reduce:generate:37: rank=1621 size=2048
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3372669133:0:1836:EmberEngine:Reduce:generate:37: rank=1836 size=2048
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3372670733:0:1654:EmberEngine:Reduce:generate:37: rank=1654 size=2048
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3372671533:0:1613:EmberEngine:Reduce:generate:37: rank=1613 size=2048
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3372673133:0:1889:EmberEngine:Reduce:generate:37: rank=1889 size=2048
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3372676333:0:1597:EmberEngine:Reduce:generate:37: rank=1597 size=2048
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3372676333:0:1744:EmberEngine:Reduce:generate:37: rank=1744 size=2048
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3372677133:0:1467:EmberEngine:Reduce:generate:37: rank=1467 size=2048
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3372677133:0:1691:EmberEngine:Reduce:generate:37: rank=1691 size=2048
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3372678733:0:1837:EmberEngine:Reduce:generate:37: rank=1837 size=2048
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3372682733:0:1872:EmberEngine:Reduce:generate:37: rank=1872 size=2048
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3372685933:0:1596:EmberEngine:Reduce:generate:37: rank=1596 size=2048
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3372687533:0:1638:EmberEngine:Reduce:generate:37: rank=1638 size=2048
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3372690733:0:1605:EmberEngine:Reduce:generate:37: rank=1605 size=2048
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3372701133:0:1469:EmberEngine:Reduce:generate:37: rank=1469 size=2048
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3372704333:0:1745:EmberEngine:Reduce:generate:37: rank=1745 size=2048
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3372710733:0:1622:EmberEngine:Reduce:generate:37: rank=1622 size=2048
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3372714733:0:1614:EmberEngine:Reduce:generate:37: rank=1614 size=2048
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3372716333:0:1890:EmberEngine:Reduce:generate:37: rank=1890 size=2048
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3372719533:0:1357:EmberEngine:Reduce:generate:37: rank=1357 size=2048
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3372720333:0:1693:EmberEngine:Reduce:generate:37: rank=1693 size=2048
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3372739533:0:1340:EmberEngine:Reduce:generate:37: rank=1340 size=2048
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3372740333:0:1841:EmberEngine:Reduce:generate:37: rank=1841 size=2048
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3372743533:0:1452:EmberEngine:Reduce:generate:37: rank=1452 size=2048
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3372744333:0:1684:EmberEngine:Reduce:generate:37: rank=1684 size=2048
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3372747533:0:1746:EmberEngine:Reduce:generate:37: rank=1746 size=2048
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3372749933:0:1840:EmberEngine:Reduce:generate:37: rank=1840 size=2048
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3372752333:0:1353:EmberEngine:Reduce:generate:37: rank=1353 size=2048
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3372753933:0:1358:EmberEngine:Reduce:generate:37: rank=1358 size=2048
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3372759533:0:1867:EmberEngine:Reduce:generate:37: rank=1867 size=2048
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3372763533:0:1705:EmberEngine:Reduce:generate:37: rank=1705 size=2048
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3372766733:0:1747:EmberEngine:Reduce:generate:37: rank=1747 size=2048
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3372784333:0:1834:EmberEngine:Reduce:generate:37: rank=1834 size=2048
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3372785933:0:1881:EmberEngine:Reduce:generate:37: rank=1881 size=2048
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3372787533:0:1686:EmberEngine:Reduce:generate:37: rank=1686 size=2048
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3372790733:0:1454:EmberEngine:Reduce:generate:37: rank=1454 size=2048
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3372796333:0:1354:EmberEngine:Reduce:generate:37: rank=1354 size=2048
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3372801133:0:1735:EmberEngine:Reduce:generate:37: rank=1735 size=2048
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3372804333:0:1748:EmberEngine:Reduce:generate:37: rank=1748 size=2048
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3372808333:0:1877:EmberEngine:Reduce:generate:37: rank=1877 size=2048
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3372811533:0:1863:EmberEngine:Reduce:generate:37: rank=1863 size=2048
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3372811533:0:1868:EmberEngine:Reduce:generate:37: rank=1868 size=2048
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3372812333:0:1847:EmberEngine:Reduce:generate:37: rank=1847 size=2048
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3372813933:0:1749:EmberEngine:Reduce:generate:37: rank=1749 size=2048
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3372816333:0:1466:EmberEngine:Reduce:generate:37: rank=1466 size=2048
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3372820333:0:1338:EmberEngine:Reduce:generate:37: rank=1338 size=2048
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3372826733:0:1696:EmberEngine:Reduce:generate:37: rank=1696 size=2048
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3372826733:0:1860:EmberEngine:Reduce:generate:37: rank=1860 size=2048
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3372829933:0:1737:EmberEngine:Reduce:generate:37: rank=1737 size=2048
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3372835533:0:1878:EmberEngine:Reduce:generate:37: rank=1878 size=2048
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3372836333:0:1873:EmberEngine:Reduce:generate:37: rank=1873 size=2048
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3372839533:0:1736:EmberEngine:Reduce:generate:37: rank=1736 size=2048
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3372845933:0:1750:EmberEngine:Reduce:generate:37: rank=1750 size=2048
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3372855533:0:1864:EmberEngine:Reduce:generate:37: rank=1864 size=2048
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3372857133:0:1856:EmberEngine:Reduce:generate:37: rank=1856 size=2048
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3372873133:0:1738:EmberEngine:Reduce:generate:37: rank=1738 size=2048
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3372875533:0:1845:EmberEngine:Reduce:generate:37: rank=1845 size=2048
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3372884333:0:1692:EmberEngine:Reduce:generate:37: rank=1692 size=2048
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3372885133:0:1844:EmberEngine:Reduce:generate:37: rank=1844 size=2048
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3372885133:0:1874:EmberEngine:Reduce:generate:37: rank=1874 size=2048
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3372893133:0:1838:EmberEngine:Reduce:generate:37: rank=1838 size=2048
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3372893933:0:1598:EmberEngine:Reduce:generate:37: rank=1598 size=2048
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3372893933:0:1699:EmberEngine:Reduce:generate:37: rank=1699 size=2048
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3372907533:0:1711:EmberEngine:Reduce:generate:37: rank=1711 size=2048
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3372917133:0:1727:EmberEngine:Reduce:generate:37: rank=1727 size=2048
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3372936333:0:1704:EmberEngine:Reduce:generate:37: rank=1704 size=2048
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3372944333:0:1739:EmberEngine:Reduce:generate:37: rank=1739 size=2048
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3372945933:0:1701:EmberEngine:Reduce:generate:37: rank=1701 size=2048
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3372947533:0:1851:EmberEngine:Reduce:generate:37: rank=1851 size=2048
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3372956333:0:1842:EmberEngine:Reduce:generate:37: rank=1842 size=2048
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3373044333:0:1707:EmberEngine:Reduce:generate:37: rank=1707 size=2048
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3373047533:0:1728:EmberEngine:Reduce:generate:37: rank=1728 size=2048
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3373061933:0:1721:EmberEngine:Reduce:generate:37: rank=1721 size=2048
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3373061933:0:1866:EmberEngine:Reduce:generate:37: rank=1866 size=2048
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3373073133:0:1858:EmberEngine:Reduce:generate:37: rank=1858 size=2048
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3373076333:0:1709:EmberEngine:Reduce:generate:37: rank=1709 size=2048
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3373093933:0:1846:EmberEngine:Reduce:generate:37: rank=1846 size=2048
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3373109133:0:1730:EmberEngine:Reduce:generate:37: rank=1730 size=2048
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3373123533:0:1712:EmberEngine:Reduce:generate:37: rank=1712 size=2048
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3373132333:0:1731:EmberEngine:Reduce:generate:37: rank=1731 size=2048
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3373161933:0:1702:EmberEngine:Reduce:generate:37: rank=1702 size=2048
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3373163533:0:1852:EmberEngine:Reduce:generate:37: rank=1852 size=2048
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3373175533:0:1732:EmberEngine:Reduce:generate:37: rank=1732 size=2048
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3373185133:0:1733:EmberEngine:Reduce:generate:37: rank=1733 size=2048
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3373185933:0:1714:EmberEngine:Reduce:generate:37: rank=1714 size=2048
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3373219533:0:1715:EmberEngine:Reduce:generate:37: rank=1715 size=2048
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3373244333:0:1720:EmberEngine:Reduce:generate:37: rank=1720 size=2048
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3373251533:0:1850:EmberEngine:Reduce:generate:37: rank=1850 size=2048
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3373253133:0:1717:EmberEngine:Reduce:generate:37: rank=1717 size=2048
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3373260333:0:1708:EmberEngine:Reduce:generate:37: rank=1708 size=2048
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3373287533:0:1710:EmberEngine:Reduce:generate:37: rank=1710 size=2048
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3373287533:0:1722:EmberEngine:Reduce:generate:37: rank=1722 size=2048
generate
3373321133:0:1723:EmberEngine:Reduce:generate:37: rank=1723 size=2048
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3373364333:0:1725:EmberEngine:Reduce:generate:37: rank=1725 size=2048
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3373373133:0:1854:EmberEngine:Reduce:generate:37: rank=1854 size=2048
generate
3373433933:0:1716:EmberEngine:Reduce:generate:37: rank=1716 size=2048
generate
3373469133:0:1718:EmberEngine:Reduce:generate:37: rank=1718 size=2048
generate
3373537133:0:1724:EmberEngine:Reduce:generate:37: rank=1724 size=2048
generate
3373584333:0:1726:EmberEngine:Reduce:generate:37: rank=1726 size=2048
Allgather: ranks 2048, loop 3, bytes 12, latency 74.190 us

```

Barrier: ranks 2048, loop 99, latency 36.555 us
7258761933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=1 messageSize=256 iterations=35
7259605133:0:1:EmberEngine:PingPong:generate:65: rank=1 size=2048
7264493933:0:32:EmberEngine:PingPong:generate:65: rank=32 size=2048
7271214733:0:1024:EmberEngine:PingPong:generate:65: rank=1024 size=2048
7273133933:0:2047:EmberEngine:PingPong:generate:65: rank=2047 size=2048
PingPong: otherRank 1, total time 63.529 us, loop 35, bufLen 256, latency 0.908 us. bandwidth 0.282076 GB/s
7322290733:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=32 messageSize=256 iterations=35
PingPong: otherRank 32, total time 159.992 us, loop 35, bufLen 256, latency 2.286 us. bandwidth 0.112006 GB/s
7482282733:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=1024 messageSize=256 iterations=35
PingPong: otherRank 1024, total time 159.992 us, loop 35, bufLen 256, latency 2.286 us. bandwidth 0.112006 GB/s
7642274733:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=2047 messageSize=256 iterations=35
PingPong: otherRank 2047, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918 GB/s
Simulation is complete, simulated time: 7.74979 ms
All PPN=2 torus runs finished. Results in: /Users/stephenrajva/Downloads/HPC-Upload 2/SST-Stimulator/MiniFE/Torus
s/PPN_2/results
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % ls -l results/*.csv
-rw-r--r--  1 stephenrajva  staff   1014703 28 Jun 06:31 results/torus_16x16_ppn2_nodes512_stats.csv
-rw-r--r--  1 stephenrajva  staff    15439 28 Jun 06:31 results/torus_2x2_ppn2_nodes8_stats.csv
-rw-r--r--  1 stephenrajva  staff   4111235 28 Jun 06:32 results/torus_32x32_ppn2_nodes2048_stats.csv
-rw-r--r--  1 stephenrajva  staff    61938 28 Jun 06:31 results/torus_4x4_ppn2_nodes32_stats.csv
-rw-r--r--  1 stephenrajva  staff   249342 28 Jun 06:31 results/torus_8x8_ppn2_nodes128_stats.csv
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST-Stimulator/MiniFE
/Torus/PPN_2"
cp ../analyze_universal.py .
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST-Stimulator/MiniFE
/Torus/PPN_2"
python3 - << 'EOF'
import re
f = "analyze_universal.py"
src = open(f).read()

src = re.sub(r'^STUDY\s*=\s*$', 'STUDY = "MiniFE_Torus_PPN2"', src, count=1, flags=re.M)
src = re.sub(r'^TOPOLOGY\s*=\s*$', 'TOPOLOGY = "torus"', src, count=1, flags=re.M)

runs = '''RUNS = [
    ("2x2", (2, 2), "results/torus_2x2_ppn2_nodes8_stats.csv"),
    ("4x4", (4, 4), "results/torus_4x4_ppn2_nodes32_stats.csv"),
    ("8x8", (8, 8), "results/torus_8x8_ppn2_nodes128_stats.csv"),
    ("16x16", (16, 16), "results/torus_16x16_ppn2_nodes512_stats.csv"),
    ("32x32", (32, 32), "results/torus_32x32_ppn2_nodes2048_stats.csv"),
]'''
src = re.sub(r'^RUNS\s*=\s*\[.*?\]', runs, src, count=1, flags=re.M | re.S)

open(f, "w").write(src)
print("Config updated. Verify below:")
print("-" * 50)
for line in src.splitlines():
    if line.startswith(("STUDY", "TOPOLOGY", "RUNS")) or 'ppn2_nodes' in line:
        print(line)
EOF
Config updated. Verify below:
-----
STUDY      = "MiniFE_Torus_PPN2"
TOPOLOGY   = "torus"
RUNS = [
    ("2x2", (2, 2), "results/torus_2x2_ppn2_nodes8_stats.csv"),
    ("4x4", (4, 4), "results/torus_4x4_ppn2_nodes32_stats.csv"),
    ("8x8", (8, 8), "results/torus_8x8_ppn2_nodes128_stats.csv"),
    ("16x16", (16, 16), "results/torus_16x16_ppn2_nodes512_stats.csv"),
    ("32x32", (32, 32), "results/torus_32x32_ppn2_nodes2048_stats.csv"),
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST-Stimulator/MiniFE
/Torus/PPN_2"
python3 analyze_universal.py
!! missing: ./results/results/torus_2x2_ppn2_nodes8_stats.csv (skipping)
!! missing: ./results/results/torus_4x4_ppn2_nodes32_stats.csv (skipping)
!! missing: ./results/results/torus_8x8_ppn2_nodes128_stats.csv (skipping)
!! missing: ./results/results/torus_16x16_ppn2_nodes512_stats.csv (skipping)
!! missing: ./results/results/torus_32x32_ppn2_nodes2048_stats.csv (skipping)

MiniFE_Torus_PPN2
Size      Latency(ns)  Hops      Thru(GB/s)  Packets    Bits      Idle%    Stalls
-----
Saved: MiniFE_Torus_PPN2_latency_ns.png
Saved: MiniFE_Torus_PPN2_hops.png
Saved: MiniFE_Torus_PPN2_throughput.png
Saved: MiniFE_Torus_PPN2_packets.png
Saved: MiniFE_Torus_PPN2_bits.png

```



```

Saved: MiniFE_Torus_PPN2_idle_pct.png
Saved: MiniFE_Torus_PPN2_stalls.png
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE
/Torus/PPN_2"
python3 - << 'EOF'
import re
f = "analyze_universal.py"
src = open(f).read()

runs = '''RUNS = [
    ("2x2", (2, 2), "torus_2x2_ppn2_nodes8_stats.csv"),
    ("4x4", (4, 4), "torus_4x4_ppn2_nodes32_stats.csv"),
    ("8x8", (8, 8), "torus_8x8_ppn2_nodes128_stats.csv"),
    ("16x16", (16, 16), "torus_16x16_ppn2_nodes512_stats.csv"),
    ("32x32", (32, 32), "torus_32x32_ppn2_nodes2048_stats.csv"),
]'''
src = re.sub(r'^RUNS\s*=\s*[\.*?\\]', runs, src, count=1, flags=re.M | re.S)
open(f, "w").write(src)

# Show how the analyzer builds the path, so we know which prefix it adds
print("=== RUNS now ===")
for line in src.splitlines():
    if 'ppn2_nodes' in line:
        print(line)
print("=== lines that reference 'results' or build the csv path ===")
for i, line in enumerate(src.splitlines(), 1):
    if 'results' in line.lower() or 'os.path.join' in line or 'STATS_DIR' in line:
        print(f"{i}: {line}")
EOF
=== RUNS now ===
    ("2x2", (2, 2), "torus_2x2_ppn2_nodes8_stats.csv"),
    ("4x4", (4, 4), "torus_4x4_ppn2_nodes32_stats.csv"),
    ("8x8", (8, 8), "torus_8x8_ppn2_nodes128_stats.csv"),
    ("16x16", (16, 16), "torus_16x16_ppn2_nodes512_stats.csv"),
    ("32x32", (32, 32), "torus_32x32_ppn2_nodes2048_stats.csv"),
=== lines that reference 'results' or build the csv path ===
20: DATA_DIR = os.environ.get("DATA_DIR", "./results")
125: path = os.path.join(DATA_DIR, fname)
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % python3 analyze_universal.py

```

```

MiniFE_Torus_PPN2
Size      Latency(ns)  Hops    Thru(GB/s)  Packets  Bits          Idle%  Stalls
-----
2x2       198.00       1.33    0.3247      6184     3588096       92.9   0
4x4       286.44       2.13    0.9961      38904    17442496      96.2   0
8x8       451.13       4.06    4.7122      258904   122329920     97.6   0
16x16     798.32       8.03    31.5980     1869720  1234056256    98.3   0
32x32     2064.99     16.02   322.8922    15119896 20018771520   98.5   0
Saved: MiniFE_Torus_PPN2_latency_ns.png
Saved: MiniFE_Torus_PPN2_hops.png
Saved: MiniFE_Torus_PPN2_throughput.png
Saved: MiniFE_Torus_PPN2_packets.png
Saved: MiniFE_Torus_PPN2_bits.png
Saved: MiniFE_Torus_PPN2_idle_pct.png
Saved: MiniFE_Torus_PPN2_stalls.png
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE
/Torus/PPN_2"
sed -i '' 's/PPN = 1/PPN = 2/g; s/PPN=1/PPN=2/g' analyze_universal.py
python3 analyze_universal.py

MiniFE_Torus_PPN2
Size      Latency(ns)  Hops    Thru(GB/s)  Packets  Bits          Idle%  Stalls
-----
2x2       198.00       1.33    0.3247      6184     3588096       92.9   0
4x4       286.44       2.13    0.9961      38904    17442496      96.2   0
8x8       451.13       4.06    4.7122      258904   122329920     97.6   0
16x16     798.32       8.03    31.5980     1869720  1234056256    98.3   0
32x32     2064.99     16.02   322.8922    15119896 20018771520   98.5   0
Saved: MiniFE_Torus_PPN2_latency_ns.png
Saved: MiniFE_Torus_PPN2_hops.png
Saved: MiniFE_Torus_PPN2_throughput.png
Saved: MiniFE_Torus_PPN2_packets.png
Saved: MiniFE_Torus_PPN2_bits.png
Saved: MiniFE_Torus_PPN2_idle_pct.png
Saved: MiniFE_Torus_PPN2_stalls.png
stephenrajva@Stephens-MacBook-Air-4 PPN_2 %

```